Binghui Wu

Ph.D. Candidate, National University of Singapore

Binghui Wu | Google Scholar | binghuiwu99 | binghuiwu99@gmail.com

Profile

Ph.D. candidate in Electrical and Computer Engineering at the National University of Singapore, expected to graduate in December 2026. My research focuses on building robust, transferable, and adaptive AI systems for dynamic and adversarial environments, with network security as a primary research testbed. My work spans adversarial machine learning, self-supervised representation learning, zero-shot generalization, multimodal and agentic AI, and reinforcement learning. I have developed red-team frameworks for robustness evaluation, retrieval-guided VLM agents with feedback memory, and large-scale representation-learning systems for security analytics. My research interests include trustworthy AI, GenAI applications, agentic AI, intelligent decision systems, and scalable machine learning for real-world applications. I am also interested in roles that bridge AI research, technical strategy, and stakeholder engagement.

Core Competencies

- **Robust and Trustworthy AI:** adversarial machine learning, red teaming, robustness evaluation, anomaly detection, and zero-shot generalization.
- **Foundation and Representation Learning:** Transformers, self-supervised pretraining, masked modeling, tokenization, sequence representation learning.
- **Multimodal and Agentic AI:** VLMs, RAG, planner-executor agents, visual reasoning, tool routing, feedback memory, and image processing.
- **Research Engineering:** Python, PyTorch, TensorFlow, OpenCV, Scikit-learn, Linux, cloud training, testbed evaluation, and reproducible experimentation.

Education

National University of Singapore Jan 2023 – Dec 2026
Ph.D. in Electrical and Computer Engineering Singapore
Research Areas: Trustworthy and Adversarial AI, Network Security, Gen AI, Representation Learning, Multimodal Agents

National University of Singapore Jan 2021 – Nov 2022
M.Eng. in Electrical and Computer Engineering Singapore
Research Area: Resource Allocation for 5G/B5G Network Slicing, Reinforcement Learning

The Chinese University of Hong Kong, Shenzhen Sep 2017 – May 2021
B.Eng. in Electronic Information Engineering Shenzhen, China
Honors: Excellent Student Leader, Excellent Student, National Encouragement Scholarship

Experience

Institute for Infocomm Research (I2R), A*STAR **Research Intern**
Retrieval-Guided Agentic VLM for Adversarial Visual Reasoning Mar 2026 – June 2026

- Developed DARKNIGHT, an agentic VLM framework for analyzing and solving heterogeneous adversarial visual challenges under local deployment, privacy, and latency constraints.
- Benchmarked eight VLM/OCR systems and developed task-aware prompting and agentic tool routing, achieving up to a 45.4% accuracy improvement on the most challenging dataset.
- Built computer vision preprocessing tools, including color segmentation, denoising, binarization, sharpening, and line removal, and identified a degeneration-recovery trade-off that motivated selective tool application.
- Designed a planner-executor architecture with structured visual reasoning, retrieval-guided strategy selection, and a closed-loop feedback mechanism that stores pass/fail outcomes for self-improvement without model retraining.

Institute for Infocomm Research (I2R), A*STAR **Research Intern**

Protocol-Compliant Adversarial Traffic Generation May 2025 – Dec 2025

– Developed a protocol-compliant red-team generator to simulate adaptive evasion against security classifiers, enabling systematic robustness testing and hard-negative generation.
– Designed target-conditioned generation and surrogate-transfer training for targeted black-box evasion.
– Validated adversarial samples through protocol analysis, TLS fingerprint preservation, and live testbed replay.

Institute for Infocomm Research (I2R), A*STAR **Research Intern**

Unified Representation Learning for Network Traffic Aug 2024 – May 2025

– Proposed UniNet, a unified Transformer-based framework for multi-granular network traffic representation learning, with Masked Feature Prediction for self-supervised embedding learning.
– Designed *T-Matrix*, a multi-granularity tokenization and embedding scheme that jointly models packet-, flow-, and session-level information in a unified representation.
– Evaluated the unified representation and encoder design across four network security and privacy tasks, achieving an average relative F1-score improvement of 18.49% in unsupervised anomaly detection.
– Built an end-to-end pipeline for PCAP parsing, multi-level feature extraction, tokenization, model training, threshold calibration, and reproducible evaluation on large-scale unlabeled traffic traces.

Acronis **Research Intern**

IoT Security Analysis Nov 2022 – Jan 2023

– Developed ZEST, an attention-based zero-shot learning framework for unseen IoT device fingerprinting.
– Designed a transformer encoder-based feature extractor to learn latent traffic representations and derive attribute vectors for unseen classes.
– Integrated a conditional variational autoencoder to generate pseudo samples for unseen devices and convert zero-shot classification into supervised learning.
– Achieved SOTA performance on real IoT traffic datasets, with strong generalization to unseen device types.

Selected Research Projects and Publications

Wu, Binghui, Levente Csikor, et al. (2026). "Darknight: Retrieval-Guided Agentic Framework for Breaking Dark Web CAPTCHA Gates". In: *IEEE eCrime 2026*.

Wu, Binghui et al. (2026). "NETGUISE: Protocol-Compliant Adversarial Traffic Generation for Early-Stage Classifier Evasion". In: *ACM SIGMETRICS 2027 (Core A*)*.

Wu, Binghui, Dinil Mon Divakaran, Levente Csikor, et al. (2026). "RECTor: Robust and Efficient Correlation Attack on Tor". In: *IEEE Communications Magazine (JCR Q1)*, pp. 1–7. DOI: 10.1109/MCOM.001.2500251.

Wu, Binghui, PC Amogh, et al. (2026). "PreNS: A Hybrid Predictive and Real-Time Resource Allocation Framework for 5G and Beyond RAN Network Slicing". In: *IEEE Transactions on Cognitive Communications and Networking (TCCN, JCR Q1)* 12, pp. 7416–7429. DOI: 10.1109/TCCN.2026.3683197.

Wu, Binghui et al. (2025). "UniNet: A Unified Multi-Granular Traffic Modeling Framework for Network Security". In: *IEEE TCCN (JCR Q1)* 12, pp. 2424–2438. DOI: 10.1109/TCCN.2025.3585170.

Wu, Binghui, Philipp Gysel, et al. (2024). "ZEST: Attention-based Zero-Shot Learning for Unseen IoT Device Classification". In: *IEEE/IFIP NOMS (Core B)*, pp. 1–9. DOI: 10.1109/NOMS59830.2024.10575079.

Additional publications in reinforcement learning, sequence modeling at IEEE HPSR and CCNC.

Teaching, Leadership, and Additional Information

– **Teaching:** NUS Teaching Assistant (2023–2025): Led tutorials and laboratory sessions for computer networks, cloud-based IoT services, optical communications, and modern computer networks.
Leadership: President of Shaw College Student Association (2019–2020): Led a student team and delivered more than ten academic, cultural, and community events with over 100 participants per event.
Interests: Badminton, Table Tennis, Reading.

– **Languages:** Chinese (native); English (professional proficiency).